

Turbulence Modelling 2020
Final Report Presentation
July 14th.

Influence of the inlet condition on the horseshoe vortex at the foot of the cylinder

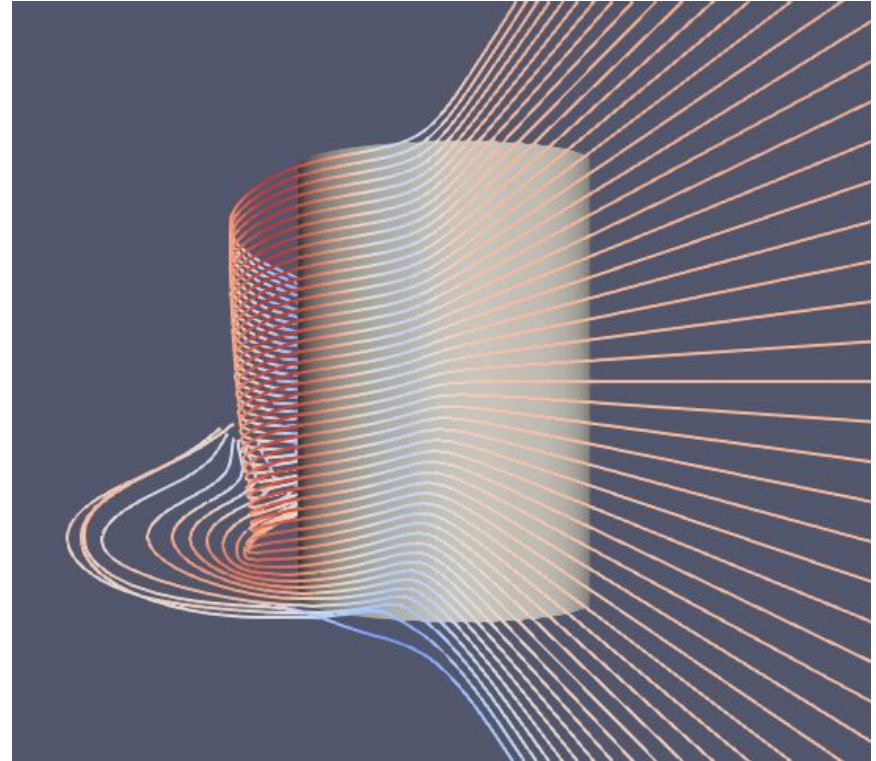
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Boundary conditions

Calculated initial condition:

$U = 0.39 \text{ m/s}$

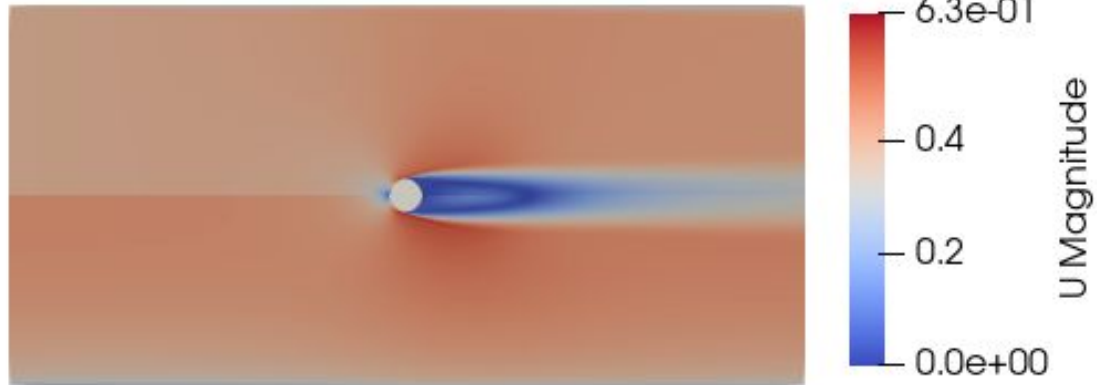
$k = 5.7038\text{e-}04 \text{ m}^2/\text{s}^2$

$\omega = 0.0262 \text{ 1/s}$

$\nu_t = 0.0020$

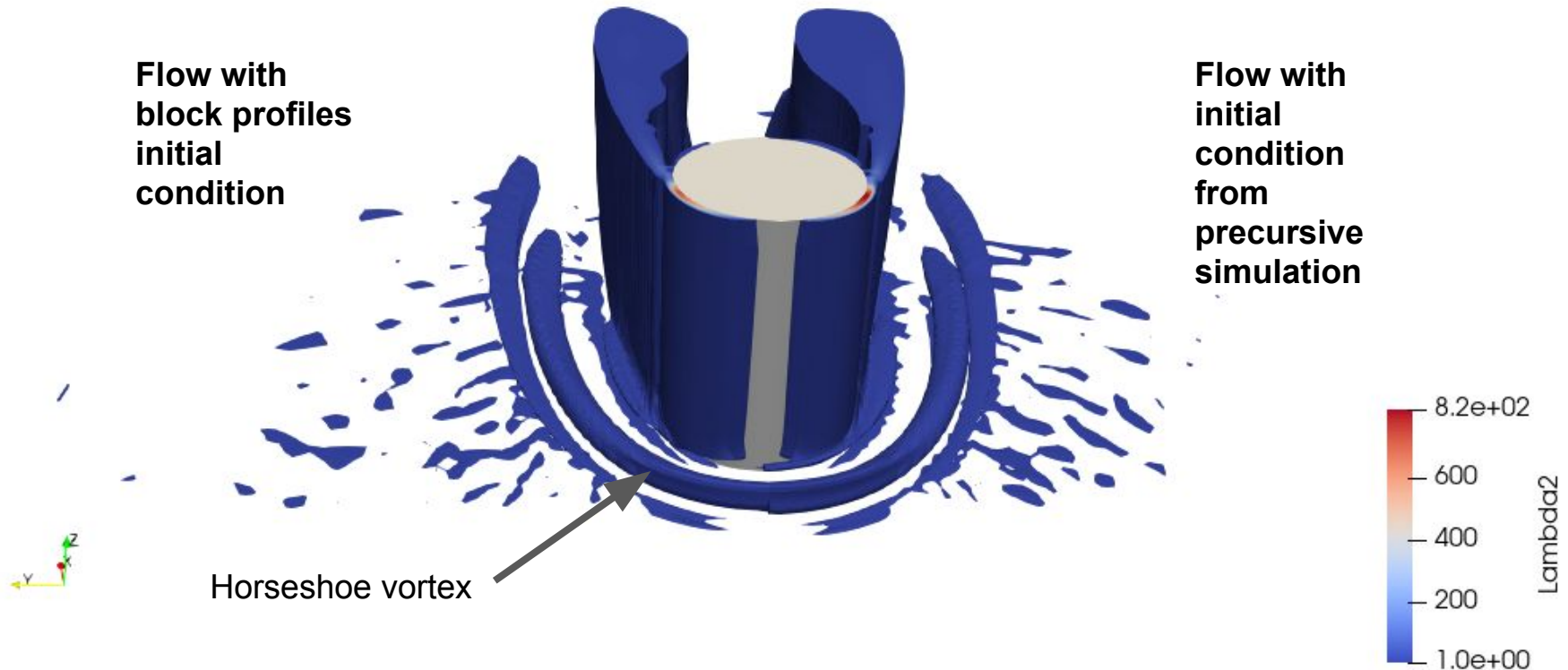
Precursive simulation:

U, p, k, ω, ν_t from
precursive channel flow

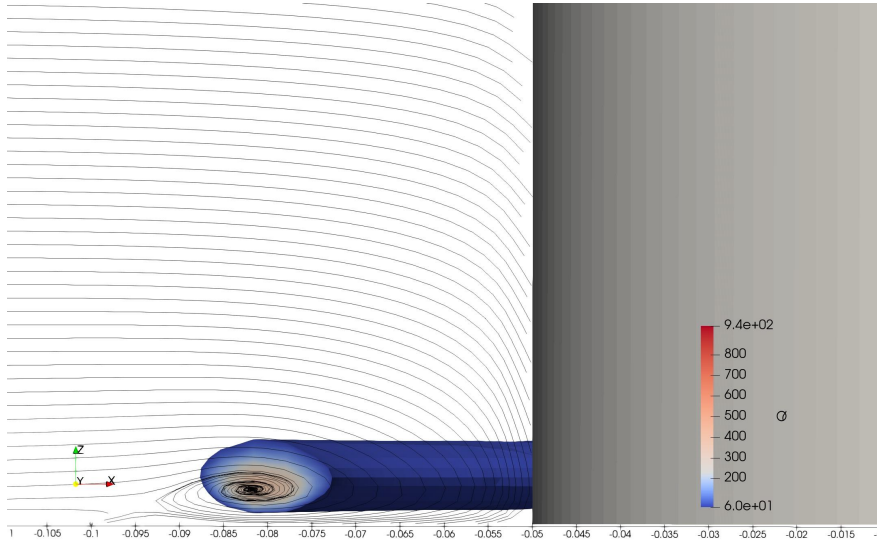


Simulation was run with
pisoFoam with $\Delta t = 0.01 \text{ s}$
for the time of 50 s.

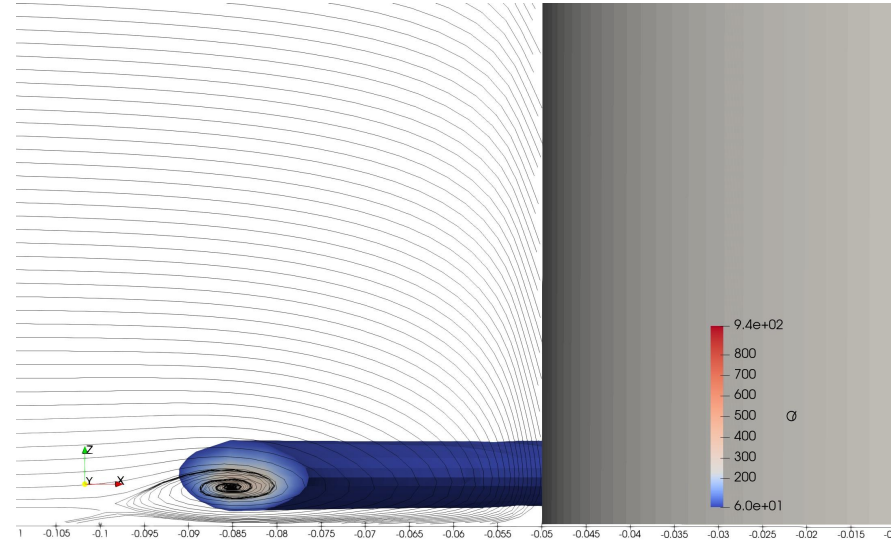
Lambda 2 criterion horseshoe vortex visualisation



Vortex in front of the cylinder



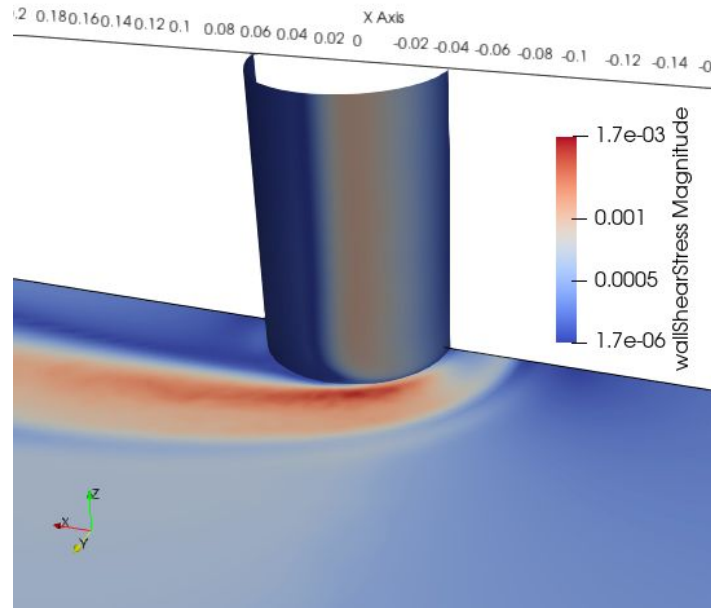
Case with block profile initial condition



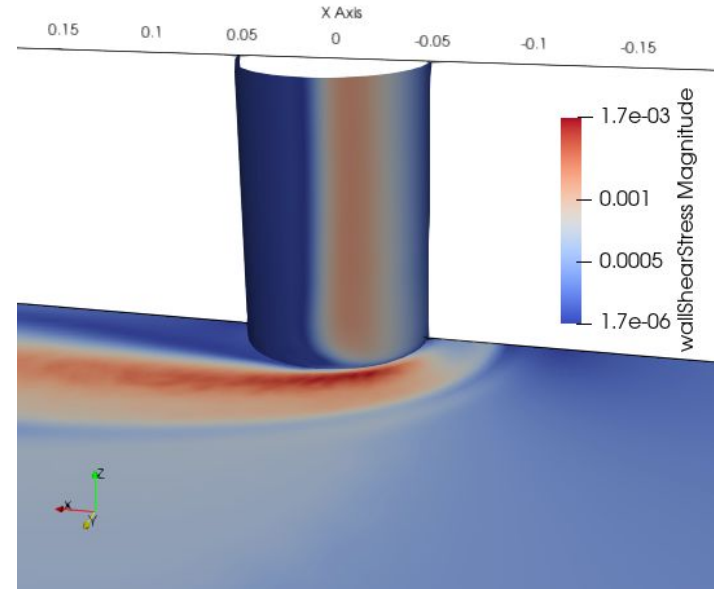
Case with precursive initial condition

In case of the precursive simulation the vortex moved to the left by $dx=0.005$ m

Flow separation at the side of the cylinder



Case with block profile initial condition



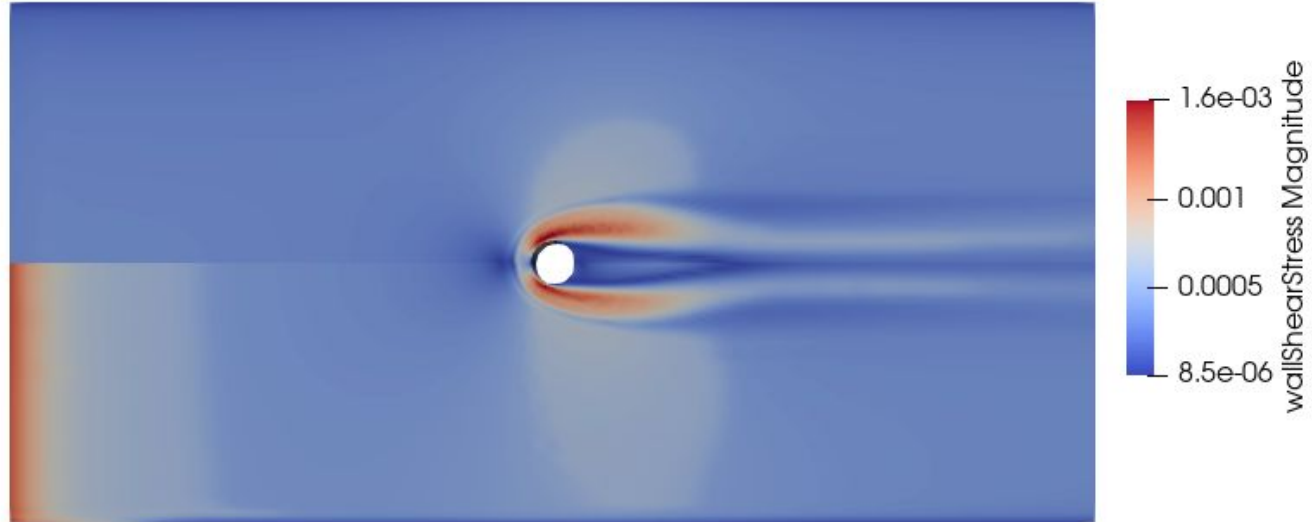
Case with precursive simulation

For both cases we observed that the detachment point at the height $z=0.075\text{m}$ is at the angle of 94.49°

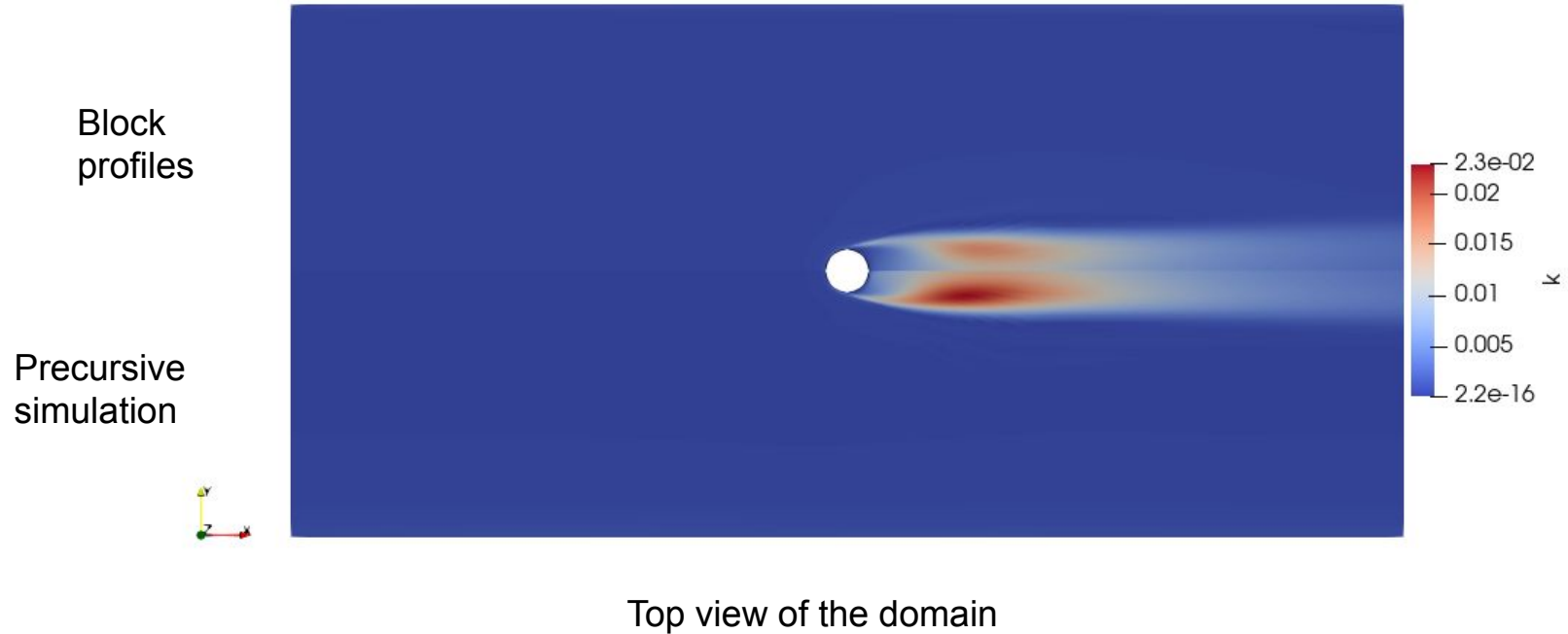
Wall shear stress footprint

Precursive
simulation initial
condition

Block profile
initial condition



Turbulent kinetic energy



Vortex identification

Q criterion - second invariant of the velocity gradient is positive in the vortex

$$Q = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_i} \frac{\partial u_j}{\partial x_j} - \frac{\partial u_i}{\partial x_j} \frac{\partial u_j}{\partial x_i} \right)$$

$$S_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

$$\Omega_{ij} = \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} - \frac{\partial u_j}{\partial x_i} \right)$$

Lambda 2 criterion - at least two eigenvalues of the tensor have to be negative

$$S^2 + \Omega^2 = S_{ij}S_{jk} + \Omega_{ij}\Omega_{jk}$$

$$\lambda_1 \geq \lambda_2 \geq \lambda_3$$

$$\lambda_2 < 0$$